Dalhousie University
CSCI 3110 — Design and Analysis of Algorithms I
Fall 2024
Assignment 3

Distributed Tuesday, September 24 2024.

Due 11:59PM, Tuesday, October 1 2024.

Instructions:

1. Before starting to work on the assignment, make sure that you have read and understood policies regarding the assignments of this course, especially the policy regarding collaboration. `https://dal.brightspace.com/d2l/le/content/338900/Home?itemIdentifier=D2L.LE.Content.ContentObject.ModuleCO-4457275`

2. To submit via Crowdmark, upload a **separate** image/PDF file (multiple pages are allowed) to each question. You may resubmit as often as necessary until the due date. A good strategy is to create an initial submission days in advance after you solve some problems, and make resubmissions later. More detailed submission information for Crowdmark can be found at:

   `https://crowdmark.com/help/completing-and-submitting-an-assignment/`

3. If you submit a joint assignment, use Crowdmark to create a group and add group members before making submissions. The "What Will Students See" section of `https://crowdmark.com/help/creating-a-group-assignment` shows the steps and screenshots of creating a group and adding members. Note that, "to avoid overwriting submissions", Crowdmark recommends "only one group member submits all the work".

4. We encourage you to typeset your solutions using LaTeX. However, you are free to use other software or submit scanned handwritten assignments as long as **they are legible**. We have the right to take marks off for illegible answers.

Questions:

1. [10 marks] Prove the following two identities using the limit method. Recall that $\lg n = \log_2 n$.

   (i) [5 marks] $200n \lg n - 10n = \Theta(n \ln n)$.

   (ii) [5 marks] $\sqrt{n}/\lg n = \omega(n^{1/3})$.

Give sufficient details in your solutions. For example, to show $\lim_{n\to\infty} \frac{\lg n}{n} = 0$, give the steps of applying l'Hopital's Rule.

2. [10 marks] We define the L_1 *distance* between two points (x_1, y_1) and (x_2, y_2) to be $|x_1 - x_2| + |y_1 - y_2|$.

   Give a set, P, of points, in which each point is colored either "red" or "blue", our task is to compute the L_1-*closest red-blue pair*, i.e., a pair of points p and q, where p is a red point and q is blue, such that the L_1-distance between p and q is minimized.

   You may assume that all coordinates are distinct.

   In this question, you are asked to solve the problem of finding the L_1-closest red-blue pair in n given points in the following special case: In this case, all red points are in region $\{(x,y)|x \geq a, y \geq b\}$ and all blue points are in $\{(x,y)|x \leq a, y \leq b\}$ for two given values a and b. Give an $O(n)$-time algorithm that computes the L_1-closest red-blue pair in this special case.

   Note: We will solve more general cases in the next assignment.

   Important: Follow the instructions given for Question 2 of Assignment 2 regarding the expectations for questions that ask you to give an algorithm for a problem. Since these requirements apply to all the assignments in this course, this reminder will not be repeated for future assignment questions.

3. [10 marks] Consider the following problem in computational geometry:

   Input: an array $Q[1..n]$ of pairs of integers (x, y) representing the x- and y-coordinates of the vertices of a convex polygon P in the plane, given in an *arbitrary order*.

   Output: an array $C[1..n]$ of pairs of integers such that $C[1], C[2], \ldots, C[n]$ gives the consecutive vertices of P in clockwise order, starting from some initial vertex.

   Give an algorithm that uses $O(n \lg n)$ time. In your algorithm, you are not allowed to call Graham's scan directly as a subroutine. Instead, borrow ideas from Graham's scan and write an algorithm that is simpler than it. Include sufficient details in your pseudocode, such as the computational steps that locate the first point in C. You can still use English words to describe the step involving sorting.